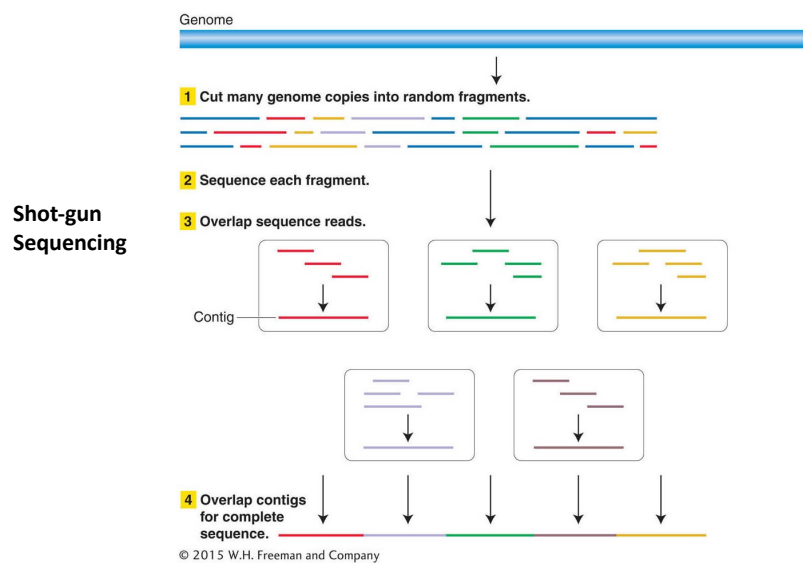


How to obtain a full genome?

- We need to “read” every single base in the genome, then put all the data back together into a completely ordered genome!
- Sequencing Technologies:
 - Sanger Sequencing
 - High-throughput sequencing, Next generation sequencing, Massively parallel sequencing:
 - Roche/454
 - Illumina Sequencing
 - Pacific Biosciences

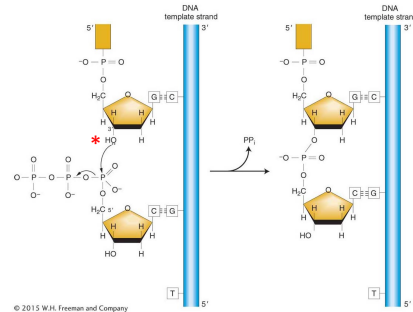
13

The genome is very big!

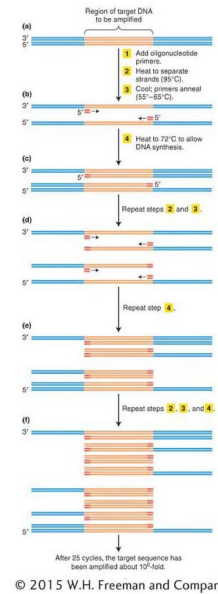


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DNA Replication and Polymerase Chain Reaction

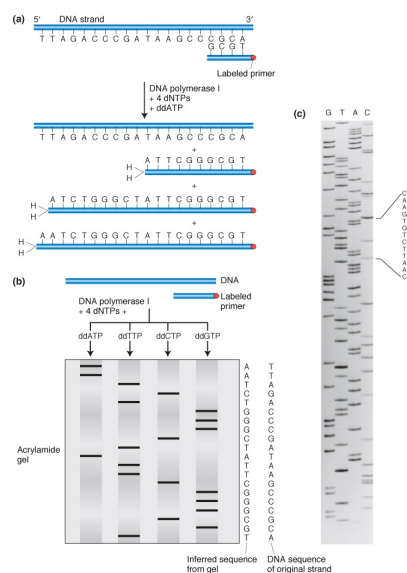


DNA synthesis involves the addition of nucleotides to a growing strand through its phosphodiester bond formation at the 3' OH group

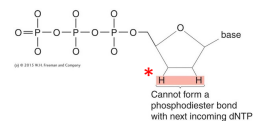


15

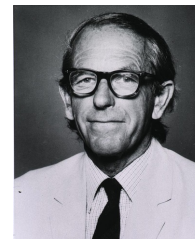
1. Sanger Sequencing dideoxy chain termination



dideoxynucleotides



* = OH in deoxy nucleotides



Frederick Sanger
Nobel Prize in Chemistry
(TWICE!)

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Automated Sanger Sequencing

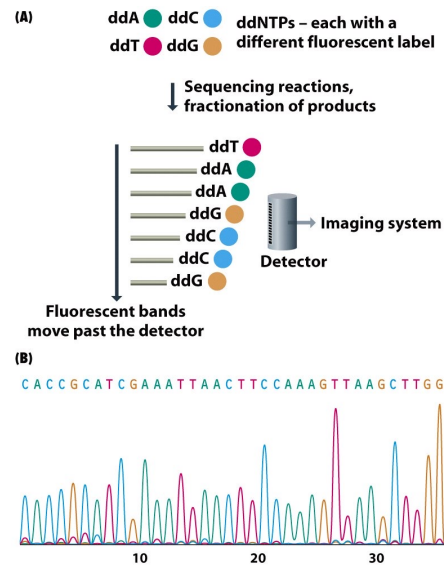
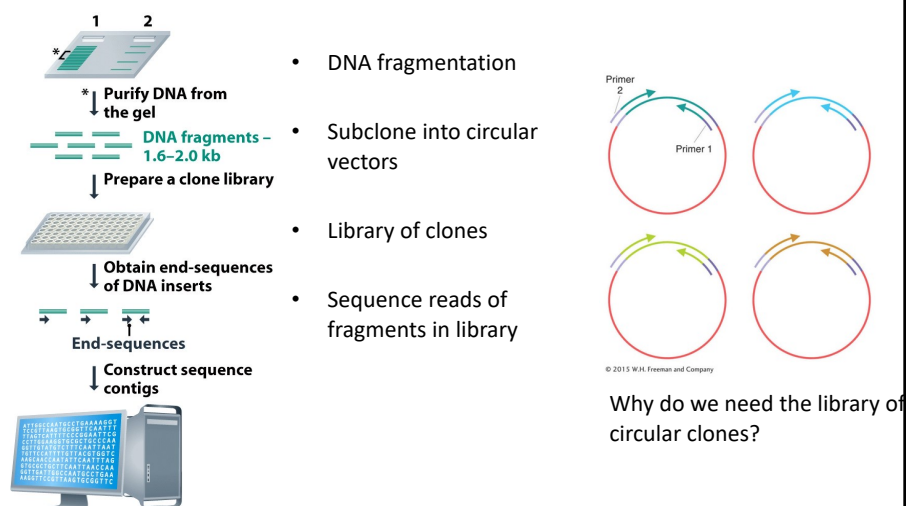


Figure 4.3 Genomes 3 (© Garland Science 2007)

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Whole genome sequencing using Sanger



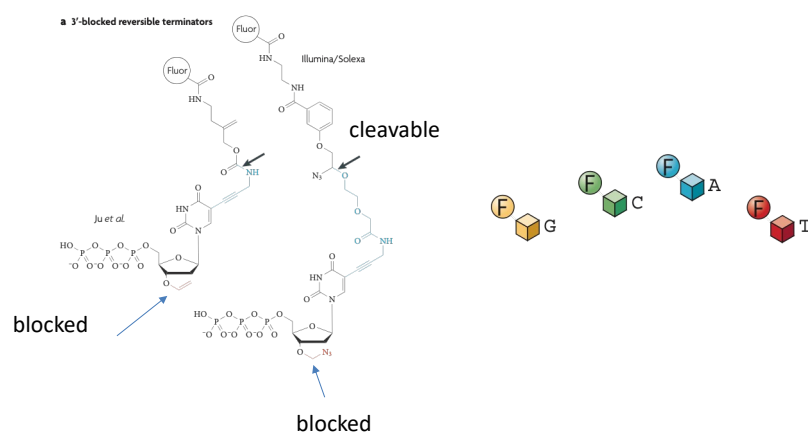
18

Whole genome sequencing using Sanger

- Accurate – low error rate
- Long reads (up to 1000 bps)
- Paired-end
- Expensive
- Slow
- Requires cloning and complicated sample preparation
- Relatively low-throughput

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2. Illumina Sequencing Sequencing By Synthesis (SBS)



- Terminated through reversible 3' terminators
- 4-color system
- Fluorescent nucleotides are imaged one at a time
- Single-strand DNA is immobilized

Mardis, 2008 &
Metzker, 2010

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2. Illumina Sequencing Sequencing By Synthesis (SBS)

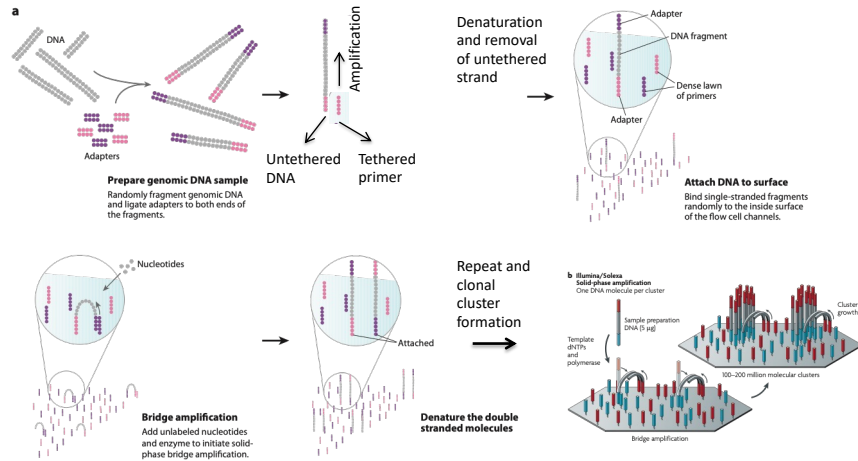


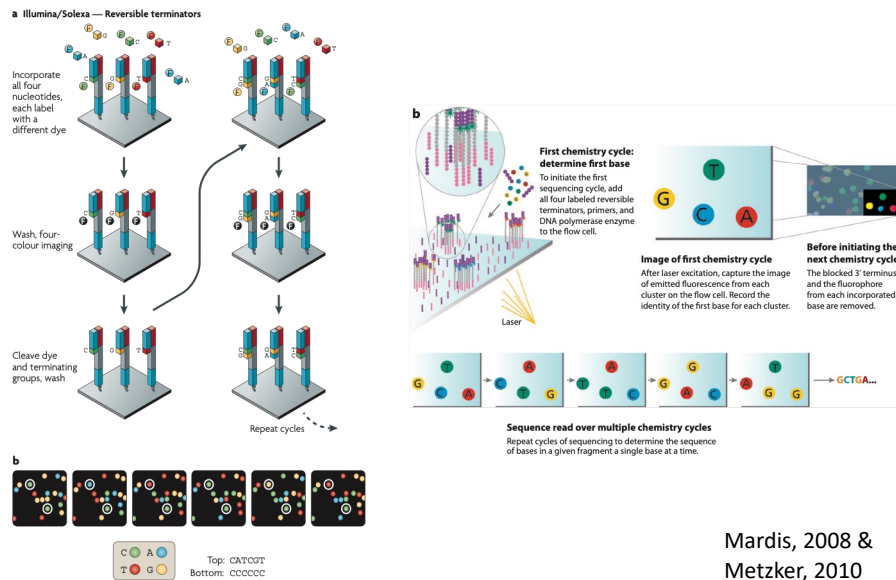
Figure 2

The Illumina sequencing-by-synthesis approach. Cluster strands created by bridge amplification are primed and all four fluorescently labeled, 3'-OH blocked nucleotides are added to the flow cell with DNA polymerase. The cluster strands are extended by one nucleotide. Following the incorporation step, the unused nucleotides and DNA polymerase molecules are washed away, a scan buffer is added to the flow cell, and the optics system scans each lane of the flow cell by imaging units called tiles. Once imaging is completed, chemicals that effect cleavage of the fluorescent labels and the 3'-OH blocking groups are added to the flow cell, which prepares the cluster strands for another round of fluorescent nucleotide incorporation.

Mardis, 2008 &
Metzker, 2010

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2. Illumina Sequencing Sequencing By Synthesis (SBS)



Mardis, 2008 &
Metzker, 2010

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- Watch this Illumina video for more details:

<https://www.youtube.com/watch?v=fCd6B5HRaZ8>